

Artificial Intelligence (AI) is the field of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, understanding natural language, and recognizing patterns.

Machine Learning (ML) is a subset of AI that enables computers to learn from data without being explicitly programmed. Instead of following fixed rules, machine learning models identify patterns in data and use those patterns to make predictions or decisions.

Deep Learning is a specialized branch of machine learning that uses artificial neural networks with multiple layers. These networks can automatically learn complex representations from large amounts of data, making them highly effective for tasks such as image recognition, speech recognition, and language translation.

Natural Language Processing (NLP) is a branch of AI that focuses on enabling computers to understand, interpret, and generate human language. Applications of NLP include chatbots, machine translation, sentiment analysis, and question-answering systems.

A Retrieval-Augmented Generation (RAG) system combines information retrieval with a large language model. Instead of relying only on the model's internal knowledge, a RAG system searches a collection of documents, retrieves the most relevant information, and provides that information to the language model as additional context before generating a response.

Embeddings are numerical representations of text. They convert words, sentences, or documents into vectors of floating-point numbers. Similar pieces of text have embeddings that are close together in vector space, allowing similarity searches to find related information efficiently.

Vector databases are designed to store and search embeddings. Popular vector databases include FAISS, ChromaDB, Qdrant, Pinecone, and Milvus. They use algorithms such as Approximate Nearest Neighbor (ANN) search to quickly retrieve vectors that are most similar to a given query.

Chunking is the process of splitting large documents into smaller pieces before creating embeddings. Proper chunking improves retrieval quality because each chunk contains a focused amount of information. Many RAG systems use chunk sizes between 300 and 1000 words with some overlap to preserve context across chunk boundaries.

When a user asks a question, the RAG pipeline first converts the question into an embedding. The vector database compares this embedding with the stored document embeddings and returns the most relevant chunks. These chunks are then included in the prompt sent to the language model, allowing it to generate accurate and context-aware responses.

Modern RAG systems often include additional components such as reranking models, metadata filtering, query expansion, and caching. These techniques improve retrieval accuracy, reduce latency, and provide more reliable answers across large document collections.